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Prediction of plant phase-separating proteins using positive-unlabeled learning

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Abstract

Liquid-liquid phase separation regulates biological processes through dynamic condensates. Despite its significance, experimentally validated phase-separating proteins in plants remain limited, complicating predictions. We overcome this gap by applying positive-unlabeled learning, a semi-supervised approach optimized for imbalanced datasets. Leveraging 6,559 reported plant phase-separating proteins from eight species, we train a model integrating sequence-structural features, enabling prediction of 174,656 high-confidence candidates across 14 species. Experimental validation confirms liquid-liquid phase separation in 67.9% of the candidate proteins from *Arabidopsis*, rice, and maize. This positive-unlabeled framework demonstrates robust predictive power while providing open resources to advance plant phase separation research.

Keywords

Plant phase-separating proteins, Positive-unlabeled learning, Semi-supervised learning framework, Multimodal features

Background

Membraneless organelles can be assembled through liquid-liquid phase separation (LLPS), a process in which specific molecules are concentrated in a liquid-like compartment that stably coexists with the surrounding liquid milieu[1, 2]. This mechanism plays a crucial role in intracellular spatial organization, participating in the regulation of various life processes[3]. In plants, proteins modulate gene expression and signal transduction via liquid-liquid phase separation, thereby exerting a key influence on growth and development and providing a unique physiological basis for adapting to environmental changes. Several representative examples illustrate the involvement of LLPS in developmental regulation. In *Arabidopsis thaliana*, FLOWERING CONTROL LOCUS A (FCA) is an RNA-binding protein (RBP) that forms condensates with FLC EXPRESSOR-LIKE 2 (FLL2, a coiled-coil protein) to concentrate RNA 3' end processing factors, thereby promoting the 3' end processing of specific RNAs and regulating flowering[4]. Similarly, during rice (*Oryza sativa*) development, the H1/H5 domain of the transcriptional repressor TELOMER REPEAT-BINDING FACTOR 2 (OsTRBF2) promotes its phase separation and repression of transcription[5]. Another rice RBP, EARLY HEADING DATE 6 (EHD6), recruits the N⁶-methyladenosine (m⁶A) reader YT521-B HOMOLOG 7 (YTH07) and sequesters *CONSTANS-LIKE 4* (*OsCOL4*) mRNA into phase-separated ribonucleoprotein condensates, thereby promoting flowering[6]. Beyond development, Phase-separating (PS) proteins are also involved in plant stress responses. The photoreceptor phytochrome B (phyB) responds to both temperature and light. Its C terminus drives phase separation through self-association, and its intrinsically disordered N-terminal extension (NTE) functions as a biophysical modulator of phase separation. Light exposure triggers a conformational change that alters phyB condensate assembly, whereas temperature sensing is directly mediated by the NTE to modulate the phase separation behavior of phyB droplets[7]. Under

heat stress, the *Arabidopsis* RBPs RNA-BINDING GLYCINE-RICH D2 (RBGD2) and RBGD4 form stress granules (SGs) through phase separation to enhance thermotolerance[8]. DROUGHT RESISTANCE GENE 9 (DRG9) is a double-stranded RNA-binding protein in rice[9]. Under drought stress, DRG9 condenses into SGs through LLPS via a crucial α -helix. DRG9 recruits the mRNAs of *NINE-CIS-EPOXYCAROTENOID DIOXYGENASE 4* (*OsNCED4*), encoding a key enzyme for the biosynthesis of abscisic acid, into SGs and protects them from degradation. LLPS mediated by RBPs plays a crucial role in the plant development and stress responses[10]. Collectively, these examples demonstrate that PS proteins contribute broadly to plant adaptation by modulating both developmental processes and stress responses.

The prediction and computational simulation of protein phase separation are potent tools for elucidating the underlying mechanisms and biological functions of this process. These methods provide theoretical forecasts that complement experimental approaches, helping elucidate the molecular mechanisms of protein function[11]. Electrostatic, hydrophobic, and π - π stacking interactions drive biological macromolecules to aggregate and undergo phase separation[12]. These interactions can occur between multiple folded domains of proteins or be mediated by intrinsically disordered regions (IDRs)[13]. The first generation of phase separation predictors, such as prion-like amino acid composition (PLAAC)[14], catGRANULE[15], and PScore[16], based on specific phase-separating driving forces, have demonstrated good predictive outcomes. Among these, PLAAC was originally used to predict prion-like domains (PrLDs). PrLDs are a special type of IDR that is rich in polar amino acids and widely involved in condensate formation[17]. For example, EARLY FLOWERING 3 (ELF3) forms condensates through its PrLD under high-temperature conditions, affecting the expression of flowering-related genes in *Arabidopsis* and leading to early flowering[18]. The current predictions of PS proteins largely rely on the primary amino acid sequence, which poses a bottleneck in predictive research, particularly when dealing with proteins that exhibit flexibility and dynamic traits. To markedly improve the accuracy and breadth of predictions, it is crucial to introduce three-dimensional (3D) structural features or other related physicochemical properties of proteins. For example, post-translational modifications play a wide range of roles in phase separation by modulating binding between interacting partners or altering the physical properties of

proteins[19]. Additionally, thermostable proteins are significantly less disordered than unstable proteins[20]. Furthermore, the structural dynamics of proteins, such as changes in their solvent accessible surface area (SASA), may alter interactions between individual proteins, thereby affecting their tendency to condense[21].

The propensity of a protein to undergo phase separation is governed by a complex set of sequence-dependent factors, necessitating the development of new predictors that integrate a broader spectrum of features associated with phase separation[22]. Machine learning (ML) approaches are robust enough to predict protein functions and can be trained on models that incorporate features related to protein LLPS. Features derived from protein sequences can work synergistically. The current mainstream approach for predicting PS proteins involves collecting experimentally validated proteins undergoing LLPS, integrating multiple features computed from protein sequence information, and then using appropriate ML models to make predictions (Fig. 1a). Based on the first generation of predictors, various predictive software packages or web servers have been developed, such as MolPhase[23], PSPredictor[24], FuzDrop[25], PhaSePred[26], Droppler[27], DeePhase[28], and PSPer[29], using ML methods like extreme gradient boosting (XGB) and multi-head neural attention. By integrating this multidimensional information, a more comprehensive understanding and prediction of protein phase separation behavior can be achieved.

Predicting PS proteins is fraught with challenges due to the scarcity of experimentally validated proteins undergoing LLPS and the low reliability of proteins purported to not form condensates. Protein LLPS prediction tools should address the following three issues: first, the insufficient number of training samples, which makes large-scale predictions difficult; second, the imbalance between positive and negative samples, with few positive samples; and third, the uncertainty of the negative samples, most of which are unknown (unlabeled). Therefore, the core issue in predicting PS plant proteins lies in how to identify positive samples from unlabeled samples.

To address the biological scenarios of predicting proteins undergoing LLPS, we explored whether a semi-supervised learning framework would raise prediction accuracy. The positive-unlabeled (PU) learning algorithm represents a sophisticated ML technique that integrates a modest amount of labeled data (i.e., experimentally validated PS proteins) and a much larger volume of unlabeled data (i.e., extensive

protein sequence databases) for model training[30]. This algorithm is adept at handling imbalanced datasets and can learn effectively from a limited number of positive samples, making it highly suitable for predicting PS proteins. It needs to be used in conjunction with a base classification algorithm, so we chose the light gradient boosting machine (LGB) algorithm as the classifier. We demonstrate here that the “PU + LGB” framework can significantly enhance prediction performance, identifying 174,656 proteins predicted to undergo LLPS with a high probability in 14 plant species. This list of proteins is publicly available at <https://www.ppsproteins.com/> for the research community to use.

Results

Training samples and predictive scenarios

We chose eight representative species of the green lineage, namely, *Arabidopsis thaliana* (At), *Brassica rapa* ssp. *pekinensis* (Br), the green alga *Chlamydomonas reinhardtii* (Cr), *Oryza sativa* (Os), the moss *Physcomitrium patens* (Pp), *Solanum lycopersicum* (Sl), *Triticum aestivum* (Ta), and *Zea mays* (Zm), for a systematic analysis of protein LLPS (Additional file 1: Table S1). For these species, we analyzed a set of 6,559 proteins that were previously identified as likely to undergo LLPS, using a cell-free system based on the precipitation of PS proteins by the small molecule biotinylated isoxazole (b-isox) [31]. In theory, the b-isox microcrystals may retrieve both constitutive and stimulus-specific PS proteins. Therefore, using this positive dataset for model training should help us identify key PS proteins involved in plant development and/or stress responses. We used the remaining proteins from each reference proteome as background proteins, designated as unlabeled samples. To eliminate data redundancy within each species' proteome, we used the CD-HIT algorithm to cluster the protein sequences,[32] resulting in a background set of 148,614 unlabeled proteins (see Methods for details). The pipeline we followed in this study is illustrated in Fig. 1a.

To assess the LLPS predictability in different species, we set up two scenarios for model training (Fig. 1b, c). We randomly extracted a number of proteins from the background set equal to that of the positive set and considered these proteins as the negative set; we performed this step 10 times to enhance statistical reliability. For inter-species predictions, we adopted a cross-validation strategy, where we

sequentially used the proteins from one species as the test set, and the positive and negative samples from the seven other species served as the training set. As a second scenario, we explored a bulked prediction method, considering all 6,559 proteins likely to undergo LLPS proteins from all eight species as one set, from which we randomly extracted 800 positive samples or 800 negative samples as the test set, with the remaining 5,759 proteins serving as the training set, without distinguishing the species of origin. This method should allow for improved accuracy by integrating PS proteins from multiple species, and enable predictions in all plant species based on these representative species.

Evaluation of prediction precision with sequence-based features

We wished to explore the features related to LLPS that are derived from protein sequences and to assess their predictive validity. To this end, we extracted seven protein sequence features that are closely related to LLPS and verified their individual significance by comparing their values for sets of PS proteins and control datasets (Fig. 1d). We used the PLAAC score to assess prion-like propensity[14], the catGRANULE score to calculate granule-formation propensity[15], the IDR score from the ESpritz algorithm[33], and the low-complexity region (LCR) score from the SEG algorithm to reflect a protein propensity to form IDRs or LCRs, respectively[34]. The values of these four features were higher for PS proteins than for the background proteins, with the PLAAC and catGRANULE scores exhibiting a large difference effect size, as determined by Cohen's d effect size. In addition to the above features, we also used the hydropathy score, the fraction of charged residues (FCR) score, and the PScore to illustrate multivalent interactions due to hydrophobicity, electrostatic interactions, and π - π stacking, respectively[16, 35]. The PScore values of PS proteins were higher than those of background proteins, with a large effect size. The hydropathy values of PS proteins were lower than those of background proteins, whereas the FCR scores showed a small difference effect size, although the differences between PS proteins and background proteins were significant.

Next, we explored the data using the LGB algorithm, performing cross-validation based on individual features and assessing the resulting prediction accuracy. All seven sequence-based features showed some predictive power (Fig. 1e and Fig. 1h), ranked in the following descending order based on the area under the receiver operating characteristic curve (AUC) values: catGRANULE (0.73), PScore

(0.71), PLAAC (0.71), hydropathy (0.68), IDR (0.62), LCR (0.58), and FCR (0.54). The predictive effectiveness of each feature was positively correlated with the size of the difference effect between the PS protein set and the background set (Fig. 1d).

Models integrating multimodal features outperform first-generation prediction software based on single features for the prediction of PS proteins[26]. To enhance our prediction accuracy, we thus employed an integrated multimodal feature strategy for cross-validation of eight species and 10-fold bulked predictions (Fig. 1f, g). We calculated the mean AUC values for inter-species predictions and bulked predictions using integrate multimodal features, obtaining values of 0.801 and 0.815, respectively, both of which were higher than the predictive accuracy of individual features (with a mean AUC value of 0.653). This result also validated the feasibility of inter-species predictions and indicated a certain degree of conservation in the region promoting LLPS of PS proteins across different species.

Given that the prediction accuracy of positive samples is crucial for this study, we specifically recorded the true positive rates for all prediction methods (Fig. 1h). Individual-feature predictions had a mean true positive rate across the seven features of 0.581, whereas inter-species prediction integrating multimodal features reached a mean true positive rate of 0.700 across the eight species, and 10-fold bulked prediction had a mean true positive rate of 0.711. These results provide a theoretical basis for further developing efficient plant PS protein prediction models. However, the prediction performance based on the LGB algorithm was not fully satisfactory, particularly regarding the accuracy of positive sample predictions.

PU learning enhances the prediction accuracy of PS proteins

In the field of bioinformatics, precise data labeling poses a significant challenge, particularly for the molecular validation of PS proteins. Due to the limitations of validation experimental equipment, calling a protein a non-PS protein is unreliable, and this protein should be considered essentially as unlabeled instead. Therefore, most of the available data are of the positive-unlabeled type. To address this issue, we introduced a PU learning algorithm in our pipeline, which allows a ML classifier to learn from a limited number of positive samples and a substantial amount of unlabeled samples, thereby improving the recognition rate (Fig. 2a). PU learning generates training sets by randomly combining positive samples and unlabeled samples and treating the unlabeled samples as negative samples. Simultaneously, PU

learning can estimate out-of-bag (OOB) unlabeled samples that are not selected through the chosen training samples. The PU learning algorithm needs to be combined with a base classification algorithm. In this study, we adopted the “PU + base classifier” approach and evaluated the prediction effects in two scenarios.

In scenario one, we used the “PU + LGB” model to conduct 10-fold bulked and inter-species predictions of PS proteins and recorded their true positive rate values (Fig. 2b) and AUC values (Fig. 2c, d). Compared to the LGB algorithm alone, adding the PU learning algorithm significantly improved the prediction accuracy of positive samples (Fig. 2b). Although the AUC value was not significantly higher (Fig. 2c, d), this result does not imply that the results were poor. The AUC value is inaccurately assessed for unlabeled samples treated as negative samples. Conversely, PU learning provides a more accurate evaluation of unlabeled samples, potentially recalling hidden positive samples within them.

Based on the above results, we designed a prediction scenario, termed the “hidden data labels” model (Fig. 2e), whereby we hid some of the PS protein labels by considering them as unlabeled background samples and calculating the resulting positive sample recall rate of different models. To assess the performance of different algorithms in recall with hidden labels, we introduced six mainstream ML algorithms: random forest (RF), LGB, decision tree (DT), support vector machine (SVM), naive bayes (NB), and multilayer perceptron-based deep neural network (DNN). The “PU + base model” approach improved both recall rate and accuracy (Fig. 2f–h), with the “PU + LGB” combination performing best, achieving a recall rate of up to 0.77, compared to 0.28 on average across the other algorithms. Additionally, although the DNN algorithm initially performed poorly in predicting PS proteins, its predictive performance significantly improved with the incorporation of the “PU + DNN” model.

Including a protein phosphorylation modification feature further improves precision

We hope to further explore new features that can enhance the accuracy of LLPS prediction based on existing features. Post-translational modifications (PTMs) play a wide range of roles in phase separation by modulating the reversibility of binding reactions between interacting partners or altering the physical properties of proteins[19]. Among various PTMs, we specifically selected phosphorylation as

the subject of analysis, primarily based on the following considerations: the addition of phosphate groups can positively or negatively control phase separation by changing protein charge[36]. This mechanism is clear and easy to quantify, making it suitable for introduction as a feature into the predictive model. To this end, we calculated the protein phosphorylation frequency (PPF, see below) and evaluated its potential in LLPS predictions.

Specifically, we collected datasets for known phosphorylation modifications for the four plant species *Arabidopsis*, rice, wheat, and maize from the eukaryotic phosphorylation site database (EPSD)[37]. We also predicted all phosphorylation sites, including those not independently experimentally validated, using NetPhos 3.1 software[38]. We then calculated the PPF of all proteins as an indicator (see Methods). Among the PS proteins, 1,728 proteins have known PPF values, and 1,709 proteins have predicted PPF values. In the background proteins, 11,351 proteins have known PPF values, and 11,351 proteins have predicted PPF values. We observed a significant difference in PPF values between PS proteins and background proteins, especially for rice proteins modified by phosphorylation (Fig. 3a, b).

We then incorporated the phosphorylation feature as a new variable into the “PU + LGB” framework, to assess the contribution of known and predicted protein phosphorylation modifications to the model decision. In the bulked prediction scenario, the mean true positive rate of individual PPF features was 0.60. When the PPF feature was added to the model comprising the seven previously mentioned sequence-based features, the true positive rate rose from 0.74 to 0.80 (Fig. 3c), and the F1 score rose from 0.76 to 0.80 (Additional file 2: Fig. S1a). In the hidden data labels scenario, the recall rate reached 0.93, and the F1 score was 0.79 (Fig. 3d). These results indicate that adding the phosphorylation feature improves the accuracy of PS protein predictions.

Protein structural features and thermostability further improve prediction precision

The solvent accessible surface area (SASA) of proteins affects protein folding and stability[21]. The SASA refers to the accessibility of the protein surface when it forms a complex (Fig. 3e). Changes in the SASA may alter the interactions between individual proteins, thus affecting their tendency to condense. Information on protein

interaction surface is therefore taken into account as a phase separation–related structural feature[39]. We batch-downloaded the structural information of proteins, both experimentally measured and predicted with AlphaFold2[40], at the AlphaFold Protein Structure Database[41]. We then calculated the relative solvent accessibility (RSA) of each protein to describe their amino acid composition (see Methods). We obtained structural information for 6,324 known PS proteins and 135,487 background proteins. We observed significant differences in RSA values between PS proteins and background proteins for five of the eight species assessed in this study (Additional file 2: Fig. S1b).

Living organisms are extremely sensitive to changes in temperature, with even minor fluctuations markedly affecting their growth, making it critical to distinguish between temperatures that are conducive to growth and those that lead to death. Thermically induced cell death is typically associated with protein denaturation. Thermostable proteins tend to be significantly less disordered than unstable proteins[20]. We thus looked into protein thermostability (PTS) as a key factor affecting phase separation characteristics. Accordingly, we compared PTS between PS proteins and background proteins for each of the eight plant species (Additional file 2: Fig. S1c). For seven of the eight species, proteins undergoing phase separation exhibited lower thermostability values compared to background proteins. This finding suggests that these proteins are more likely to become unstable at high temperature, thereby being more prone to phase separation. However, it is worth noting that the background proteins of the unicellular green alga *Chlamydomonas reinhardtii* are more thermally unstable than PS proteins. Consequently, when analyzing the RSA and PTS features, we excluded data from *C. reinhardtii* and focused on the remaining seven species. In subsequent predictive analyses, the proportion of PS proteins in the background proteins for the green alga was higher than that of non-PS proteins.

We employed the “PU + LGB” framework, incorporating the RSA feature and thermostability as new variables. As individual features, predictions based on RSA or thermostability showed a certain predictive power that was close to that obtained with the PPF (Fig. 3f, g). Adding the RSA or PTS to the seven sequence-based features enhanced prediction accuracy, with both TPR and F1 score rose to 0.77. Adding both RSA and PTS features to the seven features (9 features) increases TPR and F1 scores to 0.78. In the context of hidden data labels, the 9 features

improved the recall rate to 0.99 (Fig. 3h). In summary, adding PTM features, structural features (RSA), and thermostability to the model can improve the accuracy of PS protein prediction. Considering the incomplete status of phosphorylation data and differences among species in thermostability data, we did not incorporate these three features in the whole-proteome predictions. These three features should however be very effective for predictions of phase separation within species.

Proteome-scale prediction of PS proteins in plants

The genome-wide identification of PS proteins is an important step in the elucidation of cell regulatory mechanisms. We took the following steps to construct and evaluate proteome-scale prediction models based on the integrated multimodal feature strategy illustrated in Fig. 4a. Step 1: Construction of a reliable dataset. Due to the lack of systematic experimental validation for the existing 6,559 plant proteins, we additionally introduced 755 high-quality positive samples from the authoritative study “Comprehensive protein datasets and benchmarking for liquid-liquid phase separation studies” published on July 8, 2025, for model calibration[42]. Specifically: (1) 6,000 proteins were randomly selected from the 6,559 plant PS proteins as the “plant positive dataset”, with the remaining 559 serving as the validation set; (2) 700 proteins were randomly selected from the 755 reliable PS proteins as the “calibration positive dataset”, with the remaining 55 as the validation set, and the 700 samples were oversampled to maintain a 1:1 ratio with the plant positive dataset; (3) an equal number of proteins were randomly selected from 144,000 unlabeled proteins to form the negative sample set. Step 2: LGB model scoring. We used the LGB algorithm to predict the propensity for LLPS of 144,000 unlabeled proteins through 12 iterations, generating a scoring system to assess their likelihood of undergoing LLPS. Step 3: PU model scoring. We applied an out-of-bag strategy to construct 100 classifiers by randomly combining all positive samples and unlabeled samples to create a training set; we then used the PU bagging strategy to evaluate the probability that an unlabeled sample is a positive sample (OOB score). Step 4: Protein score stratification. We set a scoring threshold from 0 to 1.0 to calculate the true positive rate and F1 score for each protein in the validation set (Additional file 2: Fig. S2a–d), after which we ranked the prediction reliability of each predicted PS protein: the samples satisfying the

criteria of LGB score > 0.5 and OOB score > 0.9 were considered high-confidence PS proteins accounting for the 22.5% of the total proteins (Fig. 4b). By contrast, we considered samples with an LGB score < 0.1 and an OOB score < 0.7 as high-confidence non-PS proteins, accounting for 19.7% of all proteins. Step 5: we used the “PU + LGB” framework to generate robust prediction models by combining the reliable positive and negative samples based on LGB prediction scores and OOB scores. Through this series of scientifically rigorous steps, we not only constructed a robust prediction model that provides a quantitative score for each unlabeled protein, but also provided clear standards and a basis for the classification of PS proteins. The integration of the “calibration positive dataset”, which encompasses PS proteins from multiple species, with the “plant positive dataset”, which focuses on plant-specific potential PS proteins, combined with the application of the “PU + LGB” framework, collectively ensures the reliability and applicability of the model for predicting PS proteins in plant species.

To facilitate the exploration and validation of PS proteins by the LLPS research community, we deposited all predictions in a database (<https://www.ppsproteins.com/>) we developed, which offers both online browsing and batch download functions. This database includes the LLPS prediction scores of the eight species used in this study, along with predicted PS proteins from 875,892 protein sequences in additional six plant species, namely, soybean (*Glycine max*), cucumber (*Cucumis sativus*), upland cotton (*Gossypium hirsutum*), peanut (*Arachis hypogaea*), potato (*Solanum tuberosum*), and capsicum pepper (*Capsicum annuum*) (Additional file 1: Table S1). In our database, we offer 174,656 high-scoring proteins across 14 species that meet the criteria of an LGB score > 0.5 and a OOB score > 0.9 and may therefore undergo LLPS.

Physicochemical properties of predicted PS proteins

To derive insights into the physicochemical properties associated with phase separation from the predictive outcomes, we analyzed the PS proteins and non-PS proteins predicted across the eight proteomes (Fig. 4c, d). The proteins from *C. reinhardtii* showed a higher abundance of PS proteins than non-PS proteins. This finding may explain the previous observation that background proteins displayed greater thermostability in this species. As a unicellular organism in direct contact with

the environment, green algae may require more PS proteins to acclimate to their surroundings.

Of the seven sequence-based features analyzed, six exhibited trends that were consistent with and amplified from the initial validation (Fig. 4d and Fig. 1a). Specifically, non-PS proteins had higher hydropathy values, whereas PS proteins contained a greater fraction of charged residues (as seen in the FCR values). The first-generation predictors PScore, PLAAC, and catGRANULE assigned higher scores to PS proteins than to non-PS proteins, indicating a clear distinction. PS proteins had slightly higher scores for the LCR than non-PS proteins. However, the IDR values for PS proteins showed a striking pattern, with a much wider range of values for non-PS proteins than seen for PS proteins.

To investigate the specific contributions of each feature to the model outputs, we employed the SHapley Additive exPlanations (SHAP) method to interpret the whole-proteome model (Fig. 4e). SHAP is a game theory-based post-hoc explanation method that effectively clarifies various supervised learning models by quantifying the marginal contribution of each feature to the model output[43]. The order of features contributing to the model from highest to lowest importance was PScore, catGRANULE, IDR, FCR, PLAAC, hydropathy, and LCR. In terms of positive or negative effects on the model, higher values of PScore, catGRANULE, FCR, and PLAAC were associated with a greater positive influence on the model, indicating a higher likelihood of identifying a true PS protein. This result reflects the strong performance of the first-generation predictors for predictions of LLPS in plants. The IDR values exhibited a negative influence in this model, providing two insights. First, relying solely on IDR scores to assess the potential for phase separation has limitations, as IDR is not essential for protein phase separation. Second, the ESpritz algorithm is primarily based on the physicochemical properties of the constituent amino acids and the statistical features of protein sequences. If the amino acid composition of certain disordered sequences does not conform to the typical characteristics of disordered regions (for example, lacking specific polar or hydrophobic amino acids), lower IDR values may ensue, thereby negatively affecting the results.

Recall hidden labels and external data validation

In practical operations, obtaining all labels can be costly or even impossible. Therefore, we employed the PU learning method, which leverages unlabeled data to enhance the performance of learning algorithms, offering an advantage over the positive-negative (PN) type of algorithms that relies solely on fully labeled data. We randomly chose 20,000 proteins from both positive and negative datasets, with 20% allocated to the prediction set and 80% to the training set. We then evaluated the performance of the PN model and the PU+ model by incrementally adding positive samples to the unlabeled dataset (labeled as 0) (Fig. 5a). In the initial stage, the F1 scores of all models were similar; however, when the proportion of positive samples increased to nearly half of all proteins considered, the F1 score of the PN model dropped sharply. By contrast, the PU model maintained high performance until more than 90% of positive samples were added, reflecting its effectiveness in identifying positive samples from unlabeled data under conditions of limited positive samples, which is very useful for predicting plant PS proteins.

We collected 92 PS proteins from *Arabidopsis*, rice, maize, and wheat reported in published studies from 2020 to the present, which constituted the external validation set (Additional file 1: Table S2). Subsequently, we performed external validation of the model using this set and compared it with several existing software tools. First, a brief introduction to the composition of the final model (Fig. 5b). (1) Positive training set: including 700 “calibration positive samples” randomly selected from 755 reliable PS proteins[42]; 700 further randomly sampled from 6,000 “plant positive samples” randomly selected from 6,559 plant PS proteins; and 700 “predicted positive samples” randomly selected from the model prediction results. (2) Negative training set: including 1,400 “calibration negative samples” randomly selected from 2,121 reliable non-PS proteins, and 700 “predicted negative samples” randomly sampled from the prediction results. Regarding the validation set construction: (3) Positive validation set: including the remaining 55 “calibration positive samples” from the 755 reliable PS proteins; 70 randomly sampled “plant PS proteins” from the remaining 559 in the 6,559 plant PS proteins; and the aforementioned 92 published PS proteins from *Arabidopsis*, rice, maize, and wheat. (4) Negative validation set: 217 “calibration negative samples”

randomly selected from the 2,121 reliable non-PS proteins. Based on the above training set, we constructed the final prediction model for this study and scored the validation set to obtain the PUScore.

To further evaluate the model performance, we selected four excellent phase separation prediction software tools—DeePhase[28], PSPHunter[44], PSPire[39], and Seq2phase[45]—for prediction comparison on the same validation set. Among them, DeePhase employs a neural network-based language model. PSPHunter is based on the random forest algorithm and can identify key residues. Seq2phase performs feature extraction using a Transformer-based protein language model and integrates four machine learning algorithms to simultaneously predict scaffold and client proteins. PSPire uses the XGBoost algorithm and provides precomputed results for direct invocation in certain species. It should be noted that these tools were mostly developed based on data from humans, mice, yeast, and similar organisms, and have not been specifically optimized for plants. The comprehensive comparison results (Fig. 5c) show that the PUScore model proposed in this study performs best across multiple metrics, with an AUC value of 0.91 and an F1 score of 0.83, both superior to the other software. Particularly in positive sample identification, the predictive performance of PUScore is significantly improved, which is likely attributable to the advantages of the PU Bagging algorithm in positive sample inference.

To illustrate the functional distinctions of between PS and non-PS proteins, we performed a Gene Ontology (GO) term enrichment analysis using the annotation of the predicted proteins. The pathways significantly enriched in PS proteins included mRNA-related pathways, nucleic acid-related pathways, and nucleocytoplasmic transport pathways (Fig. 5d and Additional file 2: Fig. S3a, c). By contrast, non-PS proteins were significantly enriched in the oxidative phosphorylation, ribosome, and amino acid metabolism pathways, exhibiting a consistent trend of enrichment in several model plants (Fig. 5e and Additional file 2: Fig. S3b, d).

***In vivo* experimental validation of predicted LLPS proteins**

Based on the enrichment analysis of PS proteins, nucleic acids appear to play a crucial role in phase separation. Therefore, we chose the top 30 highest scoring proteins from

Arabidopsis, the top 50 highest scoring from maize, and high-scoring nucleic acid-related proteins from *Arabidopsis*, maize, and rice. We conducted *in vivo* phase separation experiments in *Nicotiana benthamiana* to test the validity of our prediction method (Additional file 1: Table S4). We successfully detected the accumulation of 53 high-confidence PS proteins from the three plant species as fusions to green fluorescent protein (Additional file 1: Table S3). We observed phase-separated dynamic condensate formation for 63% (12/19) of *Arabidopsis* proteins, 74% (14/19) of maize proteins, and 67% (10/15) of rice proteins (Fig. 6a-c and Additional file 2: Fig. S4). It is worth emphasizing that the *in vivo* assay in *N. benthamiana* employed for initial validation is prone to both false positives and false negatives, as elaborated in the subsequent discussion. The dynamic condensates we observed offer preliminary, but not conclusive, evidence for phase separation. Despite these inherent experimental limitations, we posit that the high observed success rate strongly supports the predictive power of our theoretical framework.

Notably, some high-scoring PS proteins have been previously reported to employ LLPS for their functions. Indeed, the top scoring proteins in *Arabidopsis* included ARGONAUTE1 (AGO1), FCA, and RNA helicase 6 (RH6, encoded by At2g45810), all of which have been previously reported to undergo phase separation. RH6, RH8, and RH12 play a key role in regulating membraneless nuclear dicing bodies (D-body) formation via LLPS[46]. Heat stress was shown to promote AGO1 phase separation and association with SG components[47]. Cells form SGs through phase separation under stress conditions. The formation of SGs temporarily inhibits mRNA translation, thereby protecting cells from damage and aiding in their recovery when normal conditions resume[48]. Several proteins with other important functions have not yet been associated with phase separation, but our study suggests that they undergo LLPS. These proteins include the nucleic acid-related proteins URIDYLYLTRANSFERASE 1 (URT1, involved in microRNA turnover[49]), and UBP1-ASSOCIATED PROTEIN 1a (UBA1a, contributing to the stabilization of mRNAs in the nucleus[50]) of *Arabidopsis*, and SINGLE-STRAND DNA ENDONUCLEASE1 (OsSEND1, encoded by Os08g0101600 and involved in DNA repair[51]).

Genes related to stress/environmental response include the plant YTHDF family of m⁶A-binding proteins EVOLUTIONARILY CONSERVED C-TERMINUS

(ECTs) required for basal antiviral defense, including ECT5[52]. Both ECT1 and ECT2 have been reported to undergo LLPS [53]. We also identified the maize family member ZmECT7 as a PS protein. ZmDnaJ59, a member of the heat shock protein 40 (HSP40) family, upregulated by heat stress[54], and basic Helix-Loop-Helix 166 (bHLH166, also reported as Phytochrome Rapidly Regulated 2 [PAR2]), which responds to light[55].

Proteins related to plant development included SUVH1/3-INTERACTING DNAJ DOMAIN-CONTAINING PROTEIN 3 (SDJ3) that regulates *WUSCHEL* expression in the shoot apical meristem[56], arginine-glycine-glycine repeats nuclear RBP B (RGGB, encoded by At4G17520) that regulates flowering in response to temperature[57], Os08g0416100 (similar to the RNA helicase OsRH42, which regulates panicle architecture[58]). Exploring the role of phase separation mechanisms will help us gain a deeper understanding of the functions of these proteins.

To more comprehensively capture and document the dynamics of LLPS, we utilized *in vivo* time-lapse imaging of representative proteins from *Arabidopsis thaliana* (At3g62200), maize (Zm100191891), and rice (Os02g0641800) (Additional file 3: Video. S1-3). These videos vividly demonstrate that condensates formed by all three proteins display hallmark LLPS features, including liquid-like properties such as droplet fusion and fission. Fig. 6d-f present representative key frames extracted from the videos, illustrating droplet fusion or disaggregation before, during, and after the dynamic events. Additionally, we further conducted fluorescence recovery after photobleaching (FRAP) experiments on these three proteins (Additional file 2: Fig. S5). The results showed rapid recovery of the fluorescence signal, indicating active dynamic exchange of protein molecules between the condensates and the surrounding solution, providing key evidence for their liquid-like properties.

Discussion

In this study, we conducted whole-proteome predictions for eight species from the green lineage, using three validation methods: internal data cross-validation, external data validation, and *in vivo* experiments in *N. benthamiana*. We achieved an overall

success rate for the top 53 high-scoring proteins in *Arabidopsis*, rice, and maize of 67.9% (36/53). We propose that the theoretical prediction accuracy exceeds the experimental validation rate, primarily due to the following two factors. First, there are limitations inherent to the experimental technical system. The *N. benthamiana* transient expression system, while offering advantages such as efficient operation and ease of microscopic observation, is widely used for preliminary *in vivo* screening of LLPS [4, 59]. However, proteins in this system may form non-specific aggregates due to overexpression, leading to artifacts, or false positive results. To minimize such errors as much as possible, we controlled protein expression levels by optimizing infiltration conditions, used irrelevant proteins as negative controls, and performed repeated validations along with FRAP experiments on key candidate proteins to confirm the dynamic fluidity characteristic of LLPS. Nevertheless, this heterologous expression system may still fail to fully recapitulate the precise regulatory mechanisms of certain proteins in their native cellular environments, thereby resulting in some truly phase separation-capable proteins not being effectively detected, i.e., false negative results. Second, protein phase separation exhibits significant condition dependency. Many proteins undergo LLPS that depends strongly on specific environmental conditions, including temperature, salt type and concentration, co-solutes, pH, and the volume excluded by other macromolecules[60]. For example, CYCLIN-DEPENDENT KINASE 2 (CDKC2) phosphorylates RNA polymerase II, maintaining the pace of the circadian clock in *Arabidopsis*[61]. CDKC2 localizes to nuclear speckles and associates strongly with the nuclear matrix[62]. Under heat stress (42°C), CDKC2 accumulated in enlarged nuclear speckles that increased in size after prolonged treatment, with noticeable condensates observed. ELF7 in *Arabidopsis* regulates flowering time[63]; the other flowering time-related protein FRIGIDA (FRI) forms nuclear condensates, which increased in size and number after cold exposure[64]. ELF7, an interactor of FRI, colocalizes with FRI in nuclear condensates when co-expressed in *N. benthamiana*. Thus, ELF7 is involved in condensate formation or, at the very least, can interact with FRI and co-exist within FRI condensates. The mechanism by which these condensates forms is likely phase separation. CDKC2 and ELF7 were two top-scoring proteins in *Arabidopsis* but failed to form condensates under our conditions when tested in *N.*

benthamiana. Therefore, the theoretical prediction success rate is likely to be higher than 70%.

Multivalent interactions are key drivers of protein phase separation[65, 66]. According to their underlying driving force when forming condensates, PS proteins can be categorized into two main classes: self-assembling proteins and constitutive proteins[26]. Self-assembling PS proteins spontaneously form condensates through their intrinsic multivalent interactions. These proteins typically exist in a dynamic equilibrium within cells, capable of rapidly assembling and disassembling under specific conditions. By contrast, constitutive PS proteins exist in relatively stable complexes within cells, where their multivalent interactions primarily serve to maintain the structure and function of protein complexes. The stability of these complexes is crucial for normal cellular function, particularly in response to environmental changes and in intracellular homeostasis. In our approach, based on the output of a reliable b-isox microcrystal detection method[31], we assembled a positive dataset that includes proteins with constitutive and stimulus-specific phase separation tendencies. Theoretically, this positive dataset provides a valuable resource for exploring the roles of PS proteins in plant development and stress responses more deeply.

Currently, we cannot distinguish between self-assembling proteins and constitutive proteins. There are two potential approaches to address this issue. The first approach is to train models separately for the two types of PS proteins based on their data sources. By categorizing them using known information or features and incorporating samples from diverse sources into the training set, we can enhance the generalizability of the models. The second approach involves classification based on the types of interactions. By establishing classification standards and using machine learning algorithms to perform cluster analysis on different interaction features, corresponding classification models can be built. Additionally, implementing an experimental validation and feedback mechanism will allow continuous optimization of the model based on experimental results. With the accumulation of more experimental data, we expect to integrate various experimental conditions (such as temperature, salt concentration, pH values) into future training models to more accurately predict the behavior of PS proteins under different physiological and environmental conditions. This approach will offer new perspectives into how plant

cells adjust to environmental changes through phase separation mechanisms and may reveal new biological rules and application potentials.

In biological research, we often encounter a large number of unlabeled samples. The core challenge in predicting PS proteins is accurately identifying true positives from these unlabeled samples. Semi-supervised learning leverages the information within unlabeled data to enhance model performance. By introducing the PU learning algorithm, we expanded the training dataset, improving the model's ability to recognize PS proteins and increasing accuracy through optimized sample selection strategies. For example, PU learning pre-classifies unlabeled data through clustering methods and selects samples similar to known positive samples for training, thereby helping the model better capture the features of PS proteins. To address the PU learning scenario, several adapted deep-learning algorithms have been released, such as NNPU[67], PUCNN[68], and GenPU[69]. Notably, deep learning has not yet been widely employed in PU learning scenarios using biological data. By constructing deep neural networks, models can automatically extract high-dimensional features from protein sequences, demonstrating stronger capabilities when dealing with complex biological data. This approach not only improves the predictive accuracy of PS proteins but also provides valuable clues for subsequent biological experiments.

Conclusions

In summary, this study addresses the challenge of predicting plant PS proteins by developing a “PU + LGB” machine learning framework that leverages sequence features, phosphorylation feature, structural feature and thermostability, overcoming limited experimental data. Validated *in vivo* experiments confirmed 67.9% accuracy under normal conditions, and the model predicted 174,656 high-confidence candidates across 14 plant species, compiled into an open-access database for community use. This resource advances plant LLPS research by enabling systematic exploration of PS proteins and their environmental sensing mechanisms.

Methods

Protein sequence processing and prediction strategy

Reference proteome sequences were obtained for eight green lineage species from the UniProt database (<https://www.uniprot.org/>). To ensure the quality of the dataset, the CD-HIT algorithm was employed to cluster protein sequences, using a sequence identity threshold of 0.4 as the selection criterion. After screening, a high-quality background set containing 148,614 unlabeled proteins was obtained. For the analysis of proteins from each species, an equal number of proteins was randomly selected from the background set to serve as negative samples, thus constructing a balanced negative dataset. For inter-species prediction, a cross-validation strategy was adopted: Each time, proteins from one species were selected as the test set, and the positive and negative samples from the seven other species were combined to construct the training set. For the bulked prediction method, known PS proteins from the eight species were mixed together, randomly selecting 800 positive PS proteins and non-phase-separating (non-PS) proteins to form the test set, whereas the remaining proteins served as the training set. In this pipeline, we did not distinguish the species origin of the proteins, but instead calculated the average prediction accuracy by repeating this procedure 10 times. This approach aims to enhance the generalization ability of the model and its prediction accuracy by increasing sample diversity.

Processing of protein features

Sequence-based features: In this study, seven sequence features closely related to protein liquid–liquid phase separation were considered. The hydropathy and fraction of charged residues (FCRs) of the proteins were obtained using localCIDER software with default parameters [35]. The hydropathy score is defined as the average hydropathy score of each residue on the standardized Kyte-Doolittle hydropathy scale, and the FCR score is calculated by dividing the total number of aspartic acid (D), glutamic acid (E), arginine (R), and lysine (K) residues by the length of the protein. To predict intrinsically disordered regions (IDRs) within proteins, the ESPRitz DisProt program was employed, with a decision threshold set at a 5% false positive rate [33]. Concurrently, the default parameters of the SEG local package were utilized to identify low-complexity regions (LCRs) within proteins [34]. The scores for IDR or LCR were defined by calculating the proportion of amino acids in the

corresponding regions relative to the total sequence length. The PLAAC tool provides three comprehensive scores for each protein sequence: linker region (LLR), core region, and predicted region (PRD). Given that the LLR score is more applicable in whole-proteome screening, the normalized LLR score was selected to assess the propensity for protein liquid droplet formation. Additionally, the first-generation LLPS protein prediction tools PScore and CatGRANULE were used to calculate the scores for the respective proteins. In the preliminary experiments (Fig. 1-2), missing PScore values were replaced with 0 for simplicity. In the whole-proteome prediction (Fig. 4), missing PScore values were replaced with the population mean to maintain data integrity and improve prediction accuracy.

Phosphorylation feature: Phosphorylation frequency refers to the ratio of the number of phosphorylation sites in a protein sequence to its sequence length. This metric was calculated based on the protein phosphorylation site data recorded in the EPSD database (<https://epsd.biocuckoo.cn/>). For predicted phosphorylation sites, the widely recognized software NetPhos 3.1 was employed for analysis. In the rare instances where real data are lacking or effective prediction was not feasible, these protein sequence features were supplemented with the mean values from the population.

Protein relative solvent accessibility: The Uniprot ID was used to batch download the structural information corresponding to all proteins from the AlphaFold Protein Structure Database (<https://alphafold.ebi.ac.uk/>), including both experimentally determined and AlphaFold2-predicted structures. On the protein level, the DSSP software was used to obtain the absolute accessibility values of all amino acids [70]. The relative solvent accessibility of the protein was calculated by dividing the sum of the normalized accessibility values of all amino acids by the sequence length.

Protein thermostability: The thermostability values of proteins at six temperature gradients of 40°C, 45°C, 50°C, 55°C, 60°C, and 65°C were calculated using the TemStaPro software[71], which is based on a protein language model. Subsequently, the average value was calculated to determine the protein thermostability (PTS) for each protein.

Model performance evaluation and model explanation

During the model training and evaluation procedure, a 4:1 split ratio was employed for the training and testing sets, supplemented by 10-fold cross-validation. In addition, several performance metrics of the model were calculated, including the area under the curve (AUC) value, accuracy, F1 score, precision, and recall, using scikit-learn 1.1.3[72]. SHapley Additive exPlanations (SHAP) belongs to the class of post-hoc explanation methods for models[43]. Its core idea is to compute the marginal contributions of features to the model output, thereby providing explanations for “black box models” from both global and local perspectives. SHAP constructs an additive explanation model, treating all features as “contributors”. The calculation of SHAP values in Python is implemented by the SHAP package, that is compatible with standard ML packages such as keras, tensorflow, scikit-learn, and pytorch[73]. In this study, a tree explainer “TreeExplainer” was used for the “base classifier” LGB, and a feature density scatterplot was drawn. The features were ranked based on the mean absolute SHAP values. Each point represents a sample, with redder colors indicating higher feature values and bluer colors indicating lower feature values. The positive and negative SHAP values indicate the positive or negative influence of the features.

Plants and growth conditions

Nicotiana benthamiana plants were used in this study, grown in a 24°C growth chamber under long-day conditions (16 h light/8 h darkness).

Agrobacterium-mediated expression in *N. benthamiana*

The full-length coding sequence of each candidate gene was cloned into the plant expression vector pEarlyGate104 (Additional file 1: Table S4). The resulting constructs were individually transformed into *Agrobacterium* (*Agrobacterium tumefaciens*) strain GV3101 and grown at 28°C. Positive colonies were used to inoculate LB medium for incubation overnight at 28°C. The bacterial pellet was collected by centrifugation (3000rpm, 26 °C, 15 minutes) and resuspended to an OD₆₀₀ of 1 in a buffer containing MES (pH = 5.7), MgCl₂ (1M), and acetyl eugenol (200mM). The cell suspension was incubated in the dark for 3 h before being infiltrated into the leaves of 6-week-old *N. benthamiana* plants using a needleless

syringe. The infiltrated plants were incubated for about 48 h at 24°C under a 16-h-light/8-h-dark photoperiod. The infiltrated leaves were observed under a confocal laser microscope for imaging (see below).

Confocal laser scanning microscopy imaging of plant materials

The infiltrated regions of leaves from 6-week-old *N. benthamiana* plants were excised and placed on a glass slide in a water droplet for microscopy analysis using a 63× oil microscope. Fluorescence was observed on a high-sensitivity laser confocal microscope (Zeiss LSM980+Airyscan2). The laser excitation wavelength was 514 nm (phi YFP). ZEISS ZEN 3.9 software was used for analysis. Proteins requiring specific treatment conditions as follows: for ECT5, leaves were treated with 5 mM salicylic acid (SA) for 40 minutes after 48 h of Agrobacterium infiltration, followed by observation; for AGO1, pco101966, ZmECT7, Zm103647784, and OsSEND1, leaves were incubated at 37°C for 60 minutes.

FRAP detection

FRAP experiments were performed using the Zeiss inverted laser scanning confocal microscope LSM880 with a 63x oil immersion objective. After photobleaching the condensates with a 488 nm laser at 100% intensity, recovery was recorded every 5-10 seconds. Fluorescence recovery was analyzed and curves were plotted using Zeiss ZEN 3.9 software and ImageJ.

Data availability

Details, repository information, accession numbers/dataset DOIs, and source references for the published datasets used in this study are provided in Additional file 1: Table S2. The external validation set comprised 92 phase-separating proteins curated from published studies from 2020 onward (Additional file 1: Table S2; refs. [4, 5, 7-9, 46, 47, 64, 74-131]). Protein features of the 14 species can be downloaded from <https://www.ppsproteins.com/>. The full code and training data used for the analyses are available in the GitHub (<https://github.com/furan2019/PSProteinPredict>) [132] and have been released to Zenodo (<https://doi.org/10.5281/zenodo.17617778>) [133], both under the MIT license.

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Author information

Contributions

X.W. conceived the project. X.W. and R.F. set up and directed the study. R.F. and S.J. performed the bioinformatics analysis and model development. Y.T., H.R., S.M., and J.Y. performed the experimental validation. X.W. and R.F. acquired funding. X.W., R.F., and X. H. wrote the manuscript. All authors read and/or edited the manuscript, and provided intellectual input.

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Ethics declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Andrew Cosgrove and Wenjing She were the primary editors of this article and managed its editorial process and peer review in collaboration with the rest of the editorial team. The peer-review history is available in the online version of this article.

Additional files**Additional file 1: Supplementary tables (Table S1–S4) (XLSX).**

Table S1. Phase-separating protein dataset for eight species and background protein dataset.

Table S2. Phase-separating proteins published in Arabidopsis, rice, and wheat since 2020.

Table S3. List of 53 high-scoring proteins from three plant species.

Table S4. List of primers used in this study.

Additional file 2: Supplementary figures (Fig. S1–S5) (PDF).

Fig. S1. The PPF prediction accuracy and normalized score for phase separation potential, as determined by the RSA and PTR for PS proteins and background proteins.

Fig. S2. Predictive reliability class of the LGB and OOB score.

Fig. S3. Gene Ontology term enrichment analysis of PS and non-PS proteins of rice and maize.

Fig. S4. *In vivo* phase separation experiments on high-scoring predicted PS proteins from Arabidopsis (a), maize (b) and rice (c).

Fig. S5. Fluorescence recovery after photobleaching (FRAP) of At3g62200 (a), Zm100191891 (b) and Os02g0641800 (c) condensates.

Additional file 3: Supplementary videos (Video S1–S3) (MP4).

Video S1. The droplet dynamic processes for the proteins At3g62200 in *N. benthamiana* leaf epidermal cells.

Video S2. The droplet dynamic processes for the proteins Zm100191891 in *N. benthamiana* leaf epidermal cells.

Video S3. The droplet dynamic processes for the proteins Os02g0641800 in *N. benthamiana* leaf epidermal cells.

Time-lapse imaging was acquired at 10-s intervals, showing the fusion and fission of liquid-like condensates. Total duration: 90 s. Scale bar: 10 μ m.

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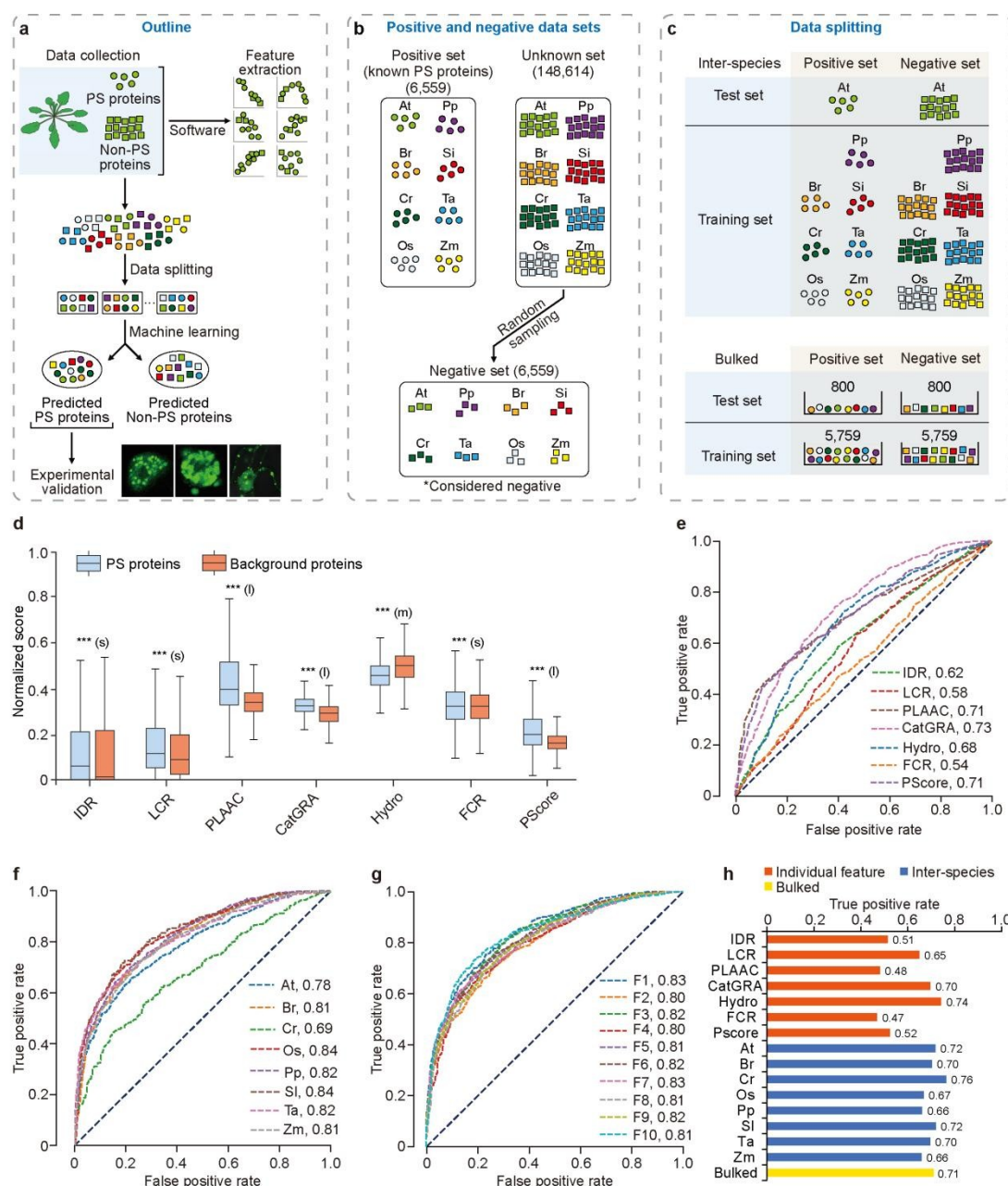
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Figure**Fig. 1 | Exploration of plant phase separation protein prediction models.**

a Diagram showing the pipeline used for the LLPS prediction model. A comprehensive collection of experimentally validated phase-separating (PS) proteins serves as the positive set. Subsequently, the sequence information from PS proteins is computed and integrated using a multi-feature approach. The dataset is then partitioned, training scenarios are established, and an appropriate machine learning model is employed to make predictions. Finally, the effectiveness of the method is assessed through experimental validation. **b** Positive and negative data sets. The positive dataset comprises 6,559 PS proteins from eight green lineage species:

Arabidopsis thaliana (At), *Brassica rapa* ssp. *pekinensis* (Br), the green alga *Chlamydomonas reinhardtii* (Cr), *Oryza sativa* (Os), the moss *Physcomitrium patens* (Pp), *Solanum lycopersicum* (Sl), *Triticum aestivum* (Ta), and *Zea mays* (Zm). The remaining 148,614 unverified reference proteins from these eight species were used as background proteins. An equal number of unlabeled proteins are then randomly selected as negative samples for training purposes. **c** Data splitting. Inter-species predictions (top panel): Proteins from one species are sequentially selected as the test set, and samples from the other seven species are used as the training set. Bulk prediction (bottom panel): Known PS proteins from all eight species are combined, and 800 PS proteins and non-PS proteins are randomly selected as the test set, with the remaining 5,759 proteins used as the training set, without distinguishing species origin. **d** Normalized score for phase separation potential, as determined by the indicated sequence-based features for PS proteins and background proteins. In the feature significance analysis, the Wilcoxon rank-sum test was used to assess the statistical significance of differences, with p -values > 0.05 , not significant (ns); *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$, and ****, $p < 0.0001$. Additionally, Cohen's d effect size was calculated to measure the actual effect size of mean differences, classifying the effect into three levels: small (s) if Cohen's $d < 0.5$, medium (m) if equal to 0.5–0.8, and large (l) if > 0.8 . **e** True and false positive rates for the indicated method, calculated as the area under the receiver operating characteristic curve (AUC) plot for individual-feature prediction based on the light gradient boosting machine (LGB) algorithm. **f** AUC plot for inter-species prediction using the integrated multimodal feature strategy. **g** AUC plot for bulked prediction using the integrated multimodal feature strategy. F1 to F10 correspond to each individual repetition of the 10-fold bulked prediction. **h** True positive rate (TPR) plot for individual-feature, inters-species, and bulked predictions based on the LGB algorithm.

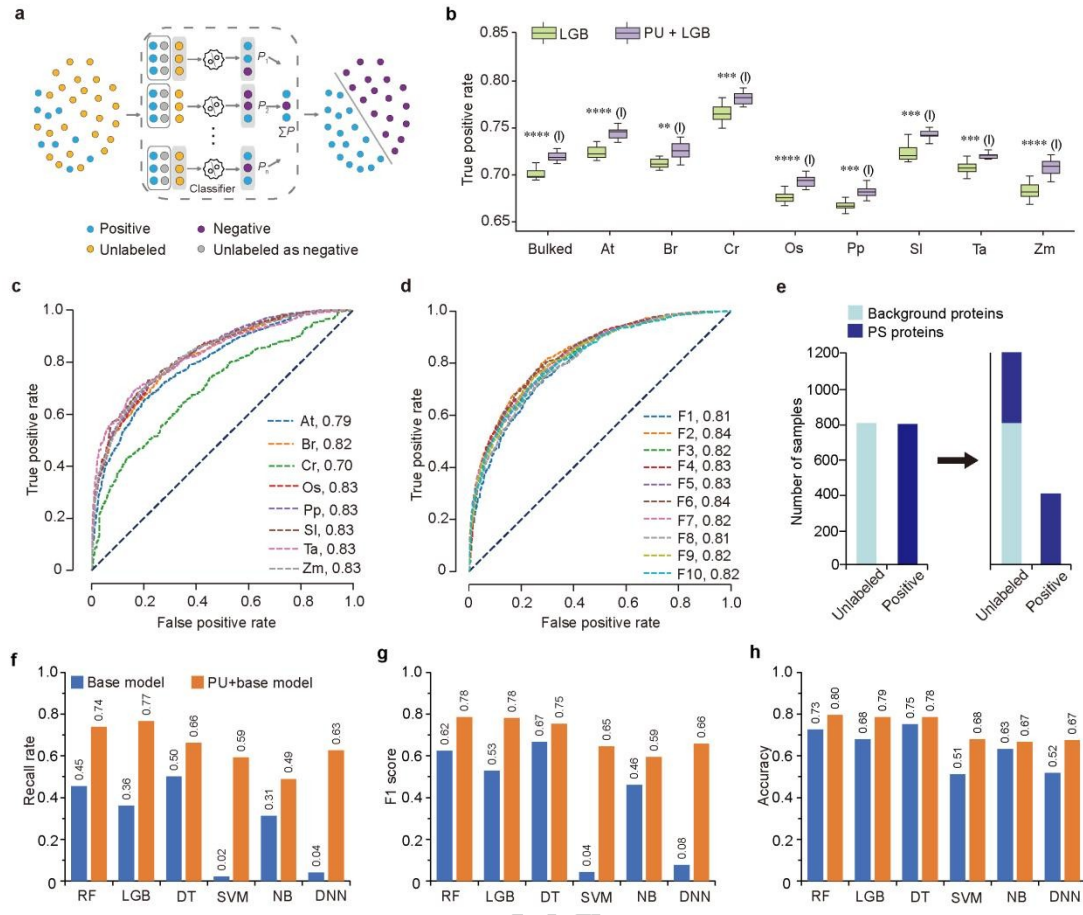


Fig. 2 | Development of the positive-unlabeled learning prediction.

a Diagram of the positive-unlabeled (PU) bagging strategy. A total of n classifiers are built, within each of which the base algorithm is used for training and prediction of positive samples, with unlabeled samples treated as negative samples. The average predicted score is calculated for each unlabeled sample to obtain its final probability of being a positive sample. **b** True positive rate plot based on the LGB algorithm alone or the “PU + LGB” algorithm. From left to right are the bulked prediction and inter-species predictions, with the species used as test set indicated. The significance of differences was assessed using a t-test, with Cohen’s d effect size used to measure the actual effect size of the mean differences: s, small; m, medium; and l, large. **c** AUC plot for inter-species predictions based on the “PU + LGB” algorithm. **d** AUC plot for the bulked prediction based on the “PU + LGB” algorithm. F1 to F10 correspond to each individual repetition of the 10-fold bulked prediction. **e** Illustration of “hidden data labels.” A total of 400 PS proteins were randomly selected, their labels were hidden, and they were added to the unlabeled background group. Predictions were conducted to assess whether different models can recall these PS proteins from the unlabeled group and how well. **f–h** Recall rate (**f**), F1 score (**g**),

and accuracy (**h**) of predicting “hidden data labels” using the indicated base models or the “PU + base models.” RF, random forest; LGB, light gradient boosting machine; DT, decision tree; SVM, support vector machine; NB, naive bayes; DNN, multilayer perceptron–based deep neural network.

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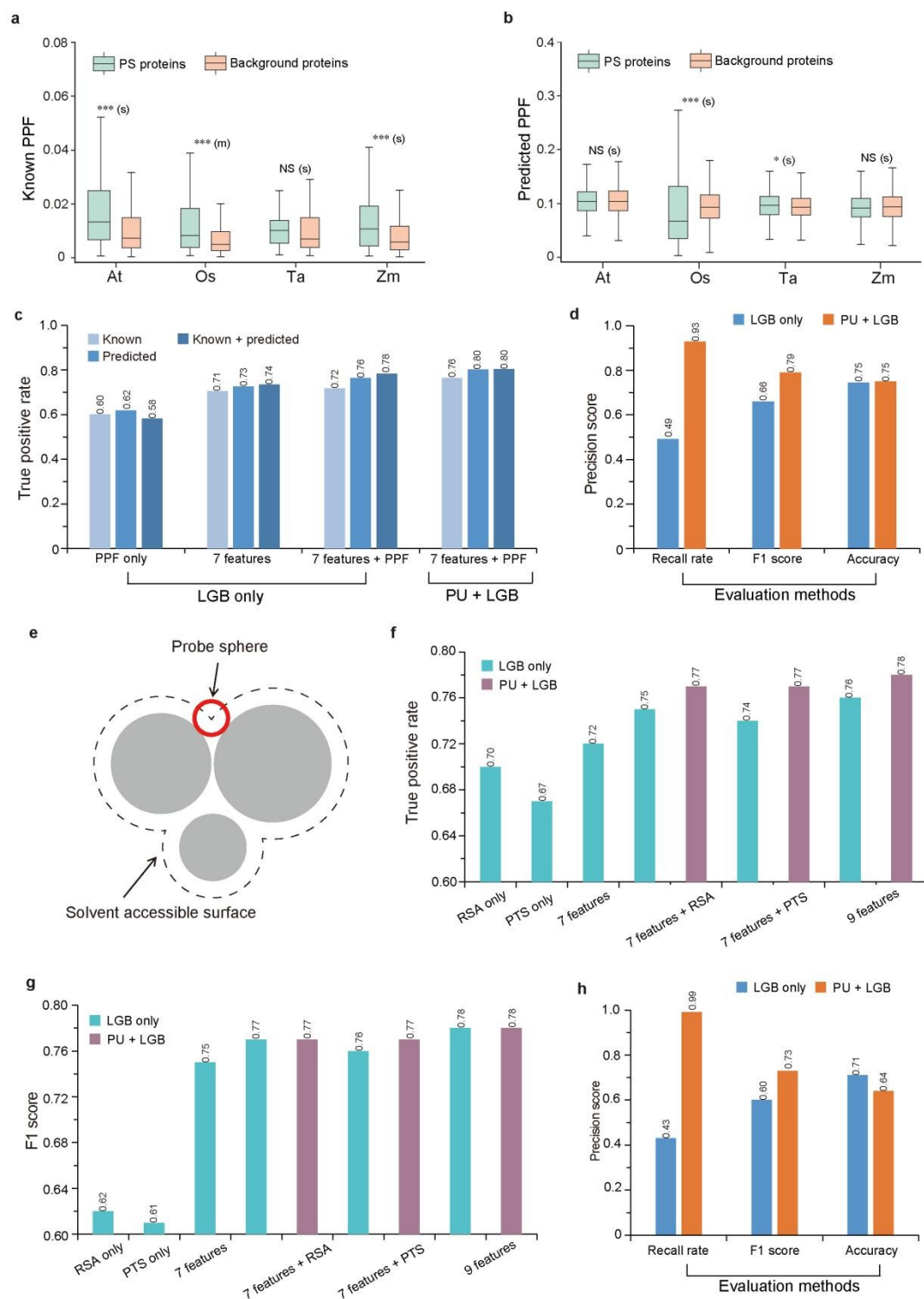


Fig. 3 | Incorporating phosphorylation frequency, relative solvent accessibility, and protein thermostability enhances precision.

a and b Boxplot showing the protein phosphorylation frequency (PPF) of PS proteins and background proteins with known (a) or predicted (b) phosphorylation sites in the indicated species. The significance of differences was assessed using the Wilcoxon

rank-sum test, and Cohen's d effect size was used to measure the actual effect size of mean differences. **c** True positive rate plot for bulked prediction scenarios using the LGB algorithm or the "PU + LGB" algorithm after adding the PPF information. The PPF alone, seven sequence-based features, or their combination were considered. **d** Using the "7 features + PPF" approach with either the LGB algorithm or the "PU + LGB" algorithm to predict PS proteins under "hidden data labels" scenarios, achieving precision metrics including recall rate, F1 score, and accuracy. **e** Illustration of the solvent accessible surface area, shown as a dashed line. The solvent accessible surface is drawn with dashed lines and is created by tracing the center of the probe sphere (in red) as it rolls along the van der Waals surface. **f and g** True positive rate (**f**) and F1 score (**g**) plots for bulked prediction scenarios using the LGB algorithm or the "PU + LGB" algorithm after adding relative solvent accessibility (RSA) or protein thermostability (PTS). Nine features, seven sequence-based features + RSA + PTS. Cyan, LGB algorithm; purple, "PU + LGB" algorithm. **h** Using the nine features approach with either the LGB algorithm or the "PU + LGB" algorithm to predict PS proteins under "hidden data labels" scenarios, achieving precision metrics including recall rate, F1 score, and accuracy.

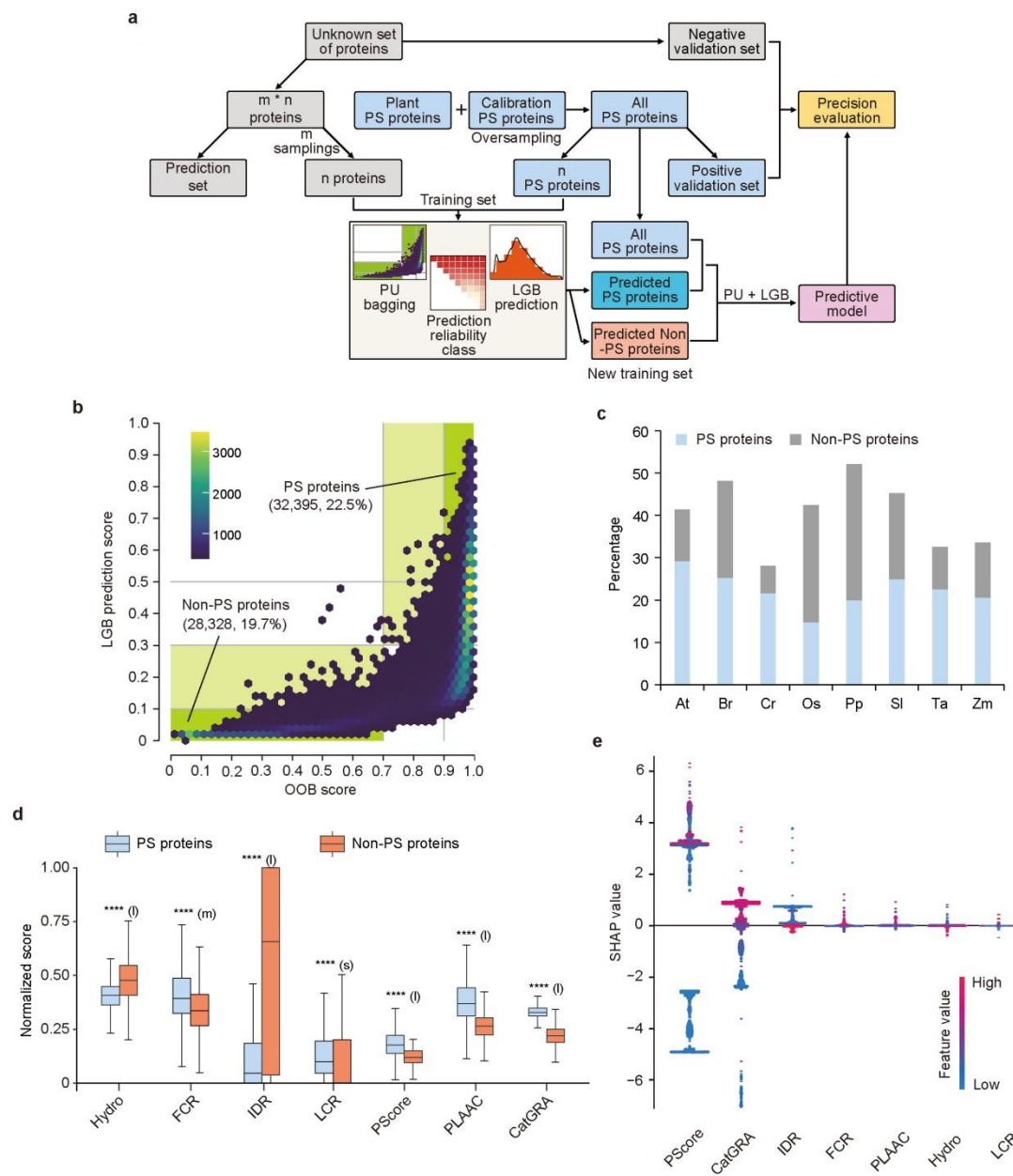


Fig. 4 | Whole-proteome scale prediction of PS proteins.

a Diagram of the whole-proteome prediction pipeline. Unlabeled samples equal in number to the positive samples were sequentially drawn to train the model, whereas the remaining unlabeled samples were used for prediction. After 12 iterations, all unlabeled samples were drawn once and predicted 12 times. The average LGB score was obtained. The PU bagging strategy was used to estimate the out-of-bag (OOB) score for unlabeled samples. The LGB and OOB scores were ranked for their predictive reliability separately. Using the reliable positive and negative samples, a robust prediction model for the “PU + LGB” machine learning framework was constructed. **b** Scatterplot of the LGB prediction score and OOB score. Samples with “LGB scores > 0.5 and OOB scores > 0.9” were considered reliable positive samples,

whereas those with “LGB scores < 0.1 and OOB scores < 0.7 ” are considered reliable negative samples. **c** Proportions of high-confidence predicted PS and high-confidence non-PS proteins in each of the eight plant species. **d** Normalized score for PS proteins and non-PS proteins based on the indicated sequence-based features. In the feature significance analysis, the Wilcoxon rank-sum test was used to assess the statistical significance of differences, and Cohen’s d effect size was used to measure the actual effect size of mean differences. **e** Plot of SHapley Additive exPlanations (SHAP) values each sequence-based feature.

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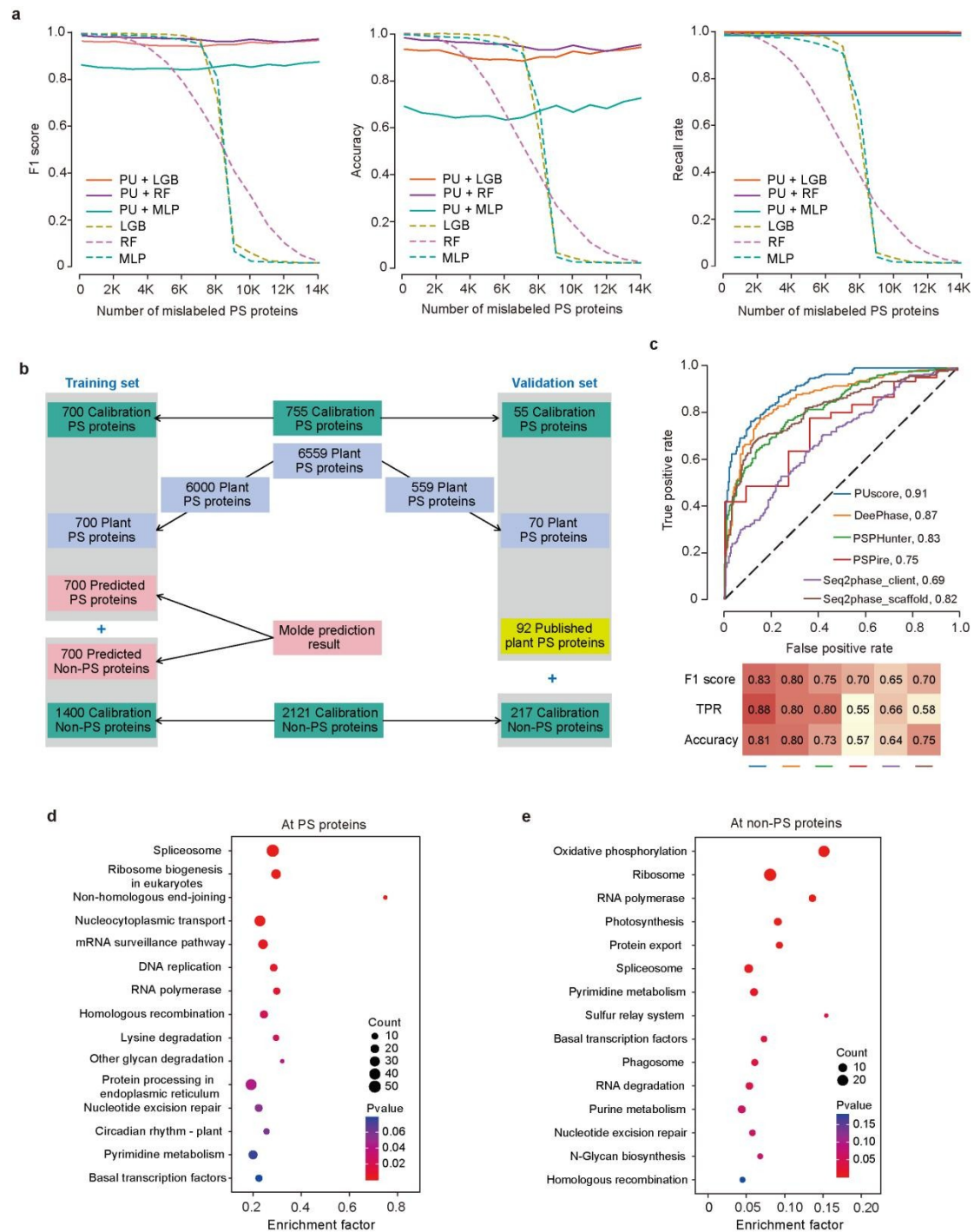


Fig. 5 | Data validation and protein function enrichment analysis.

a F1 score, accuracy, and recall rate based on PN models and PU models when PS protein labels are hidden. **b** Composition of the model. The left panel illustrates the training set, while the right panel shows the validation set. The validation set includes 92 PS proteins collected from *Arabidopsis*, rice, and wheat, sourced from studies published between 2020 and the present. **c** Performance comparison of multiple prediction tools. Our model was used to score the validation set, yielding the PUscore. This was compared with predictions from

four other tools: DeePhase, PSPHunter, PSPire, and Seq2phase. The evaluated performance metrics include the AUC value, F1 score, TPR value, and accuracy value. **d and e** Gene Ontology (GO) term enrichment analysis of PS and non-PS proteins in *Arabidopsis*. The size of the circles reflects the number of proteins, and their color indicates the *p*-value.

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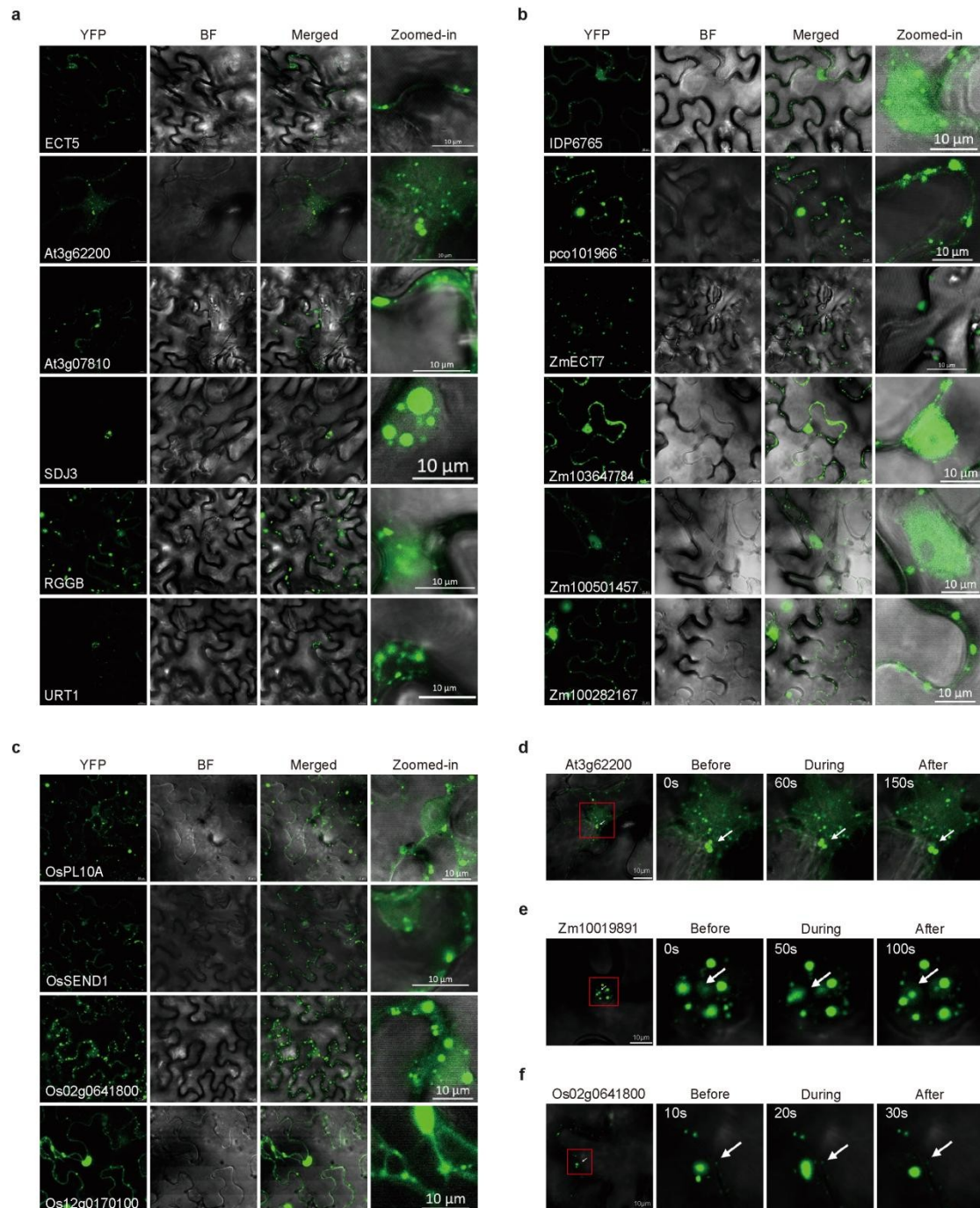


Fig. 6 | *In vivo* phase separation experiments.

a Representative images of condensate formation for selected *Arabidopsis* proteins in *N. benthamiana* leaves: ECT5, At3g62200, At3g07810, SDJ3, RGGB, URT1.

b Representative images of condensate formation for selected maize proteins in *N. benthamiana* leaves: IDP6765, pco101966, ZmECT7, Zm103647784, Zm100501457, Zm100282167.

c Representative images of condensate formation for selected rice proteins in *N. benthamiana* leaves: OsPL10A, OsSEND1, Os02g0641800, and Os12g0170100. Scale bars are shown in the last image of

each protein. BF, bright field. **d, e, and f** Representative *in vivo* droplet dynamics of PS proteins in plants, including At3g62200 in *Arabidopsis*, Zm100191891 in maize, and Os02g0641800 in rice. Representative time-lapse frames extracted from *in vivo* imaging demonstrate characteristic liquid-like behaviors of the condensates formed by these proteins, including droplet fusion and disaggregation. For each protein, sequential images corresponding to stages before, during, and after the dynamic events are shown, with the corresponding time points indicated. Scale bars are shown in the first image of each protein.

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